Cross-Validation

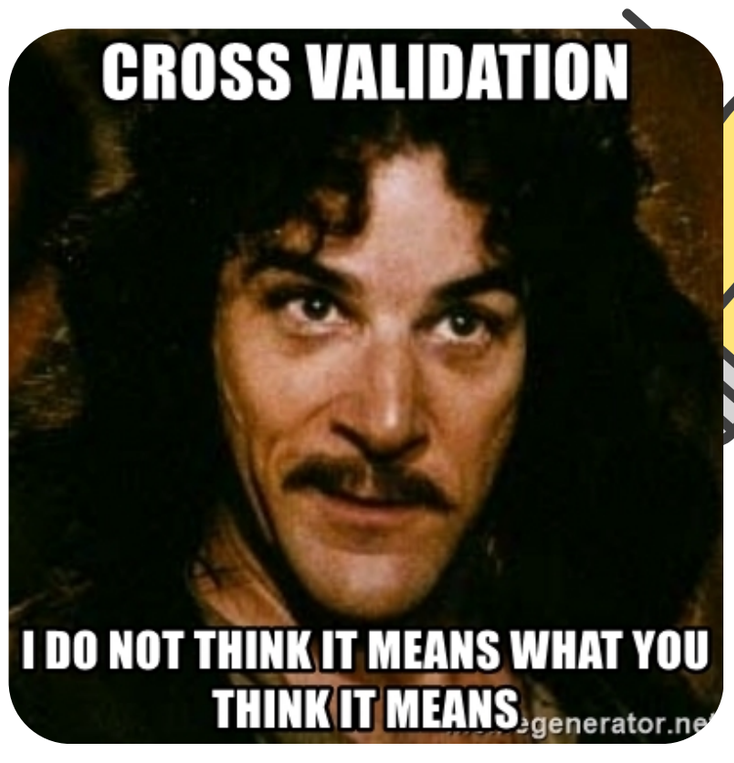
When training machine learning models, it’s not enough to just fit a model and check its performance on the same data. We need a way to estimate how well the model will perform on new, unseen data — and that's exactly where cross-validation comes in.

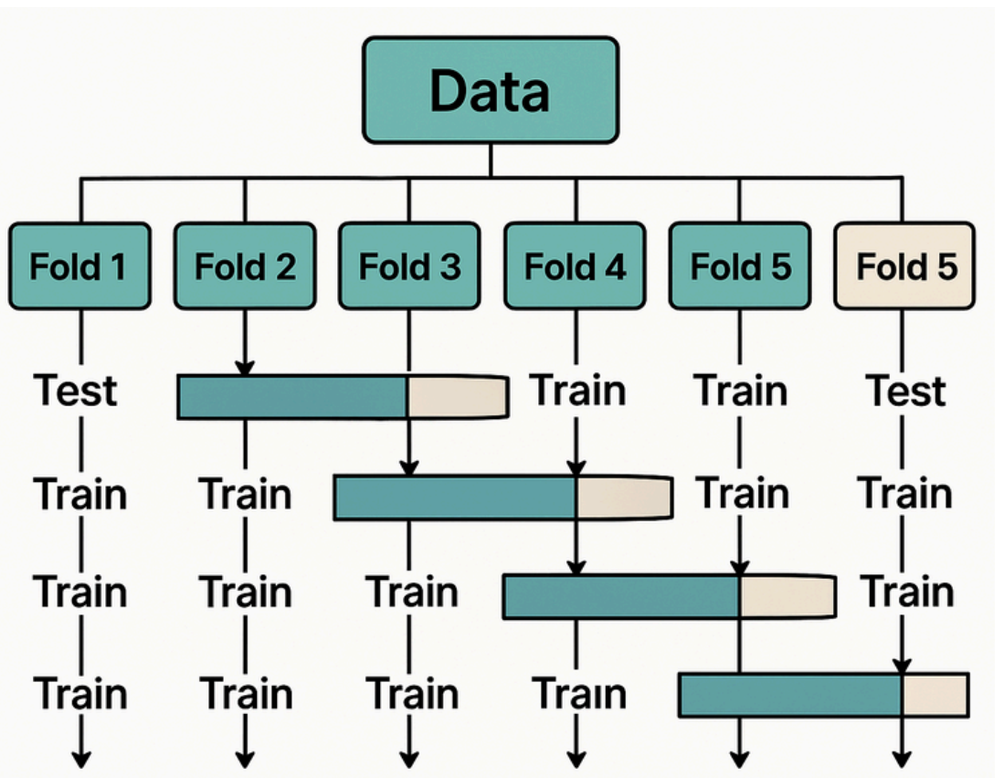
Cross-validation is a method that helps us evaluate how well a model generalizes. Instead of training the model once and testing it on one fixed validation set, we split the data into multiple parts, train the model several times, and take the average of the results. This gives us a more reliable estimate of the model’s performance.

The most common version is K-Fold Cross-Validation. In this method, the dataset is split into K equal parts (called “folds”). The model is trained K times, each time using K−1 folds for training and the remaining one for testing. This way, every data point gets to be in the test set once, and the final score is the average of all the test scores.

```python
from sklearn.model_selection
import cross_val_score

scores = cross_val_score(model,
X, y, cv=5)
print("Mean CV score:., scores
mean())
```





This method is especially useful when you don’t have a lot of data. It allows you to make the most of your dataset without having to sacrifice a big chunk of it for testing. It also helps detect issues like overfitting or underfitting more reliably.

There are other versions too. For example, Stratified K-Fold makes sure each fold has a similar class distribution — useful in classification problems. There’s also Leave-One-Out Cross-Validation (LOOCV), where each fold contains just one data point. That gives a very thorough evaluation, but it’s slow and usually not practical for large datasets.

In real projects, cross-validation helps us choose the right model, compare algorithms fairly, and tune hyperparameters. It’s an essential tool for making decisions that are based on solid evidence, not just random lucky splits.

In short, cross-validation is like doing multiple mini-experiments with your data and trusting the average result more than any single test.